

# **UMC 0.11um LOGIC/MIXED\_MODE AI Advanced Enhancement 1.2V/3.3V Diode SPICE Model Document**

**-Monte Carlo-**

**Version 0.4 Phase 1  
2017/10/20**

## Table of Contents

|          |                                                                   |           |
|----------|-------------------------------------------------------------------|-----------|
| <b>1</b> | <b>INTRODUCTION .....</b>                                         | <b>5</b>  |
| 1.1      | Revision History .....                                            | 5         |
| 1.2      | General Information.....                                          | 6         |
| 1.3      | Model Usage .....                                                 | 6         |
| 1.3.1    | The combination of statistical model & corner model .....         | 6         |
| 1.3.2    | Sigma number for Monte-Carlo model.....                           | 7         |
| <b>2</b> | <b>MONTE CARLO SIMULATION RESULTS FOR PROCESS VARIATION .....</b> | <b>10</b> |
| 2.1      | Simulation result plots .....                                     | 10        |
| 2.1.1    | 1.2V DION simulation results: .....                               | 10        |
| 2.1.2    | 1.2V DIOP simulation results:.....                                | 14        |
| 2.1.3    | 3.3V DION simulation results: .....                               | 18        |
| 2.1.4    | 3.3V DIOP simulation results:.....                                | 22        |
| 2.1.5    | 1.2V DIONW simulation results: .....                              | 26        |
| 2.1.6    | 1.2V DIODNW simulation results: .....                             | 28        |
| 2.1.7    | 1.2V DIOPW simulation results: .....                              | 32        |
| <b>3</b> | <b>APPENDIX.....</b>                                              | <b>34</b> |
| 3.1      | HSPICE Warning Message.....                                       | 34        |
| 3.2      | ELDO Warning Message.....                                         | 34        |
| 3.3      | SPECTRE Warning Message .....                                     | 34        |

## List of Figures

|                                                                                  |    |
|----------------------------------------------------------------------------------|----|
| Figure 2-1 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V" .....         | 10 |
| Figure 2-2 Device 2: Monte Carlo graph of "Capacitance @ Vbias=0V" .....         | 11 |
| Figure 2-3 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V" .....  | 12 |
| Figure 2-4 Device 2: Monte Carlo graph of "Forward Current @ Vbias= 1.0V" .....  | 13 |
| Figure 2-5 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V" .....         | 14 |
| Figure 2-6 Device 2: Monte Carlo graph of "Capacitance @ Vbias=0V" .....         | 15 |
| Figure 2-7 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V" .....  | 16 |
| Figure 2-8 Device 2: Monte Carlo graph of "Forward Current @ Vbias= 1.0V" .....  | 17 |
| Figure 2-9 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V" .....         | 18 |
| Figure 2-10 Device 2: Monte Carlo graph of "Capacitance @ Vbias=0V" .....        | 19 |
| Figure 2-11 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V" ..... | 20 |
| Figure 2-12 Device 2: Monte Carlo graph of "Forward Current @ Vbias= 1.0V" ..... | 21 |
| Figure 2-13 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V" .....        | 22 |
| Figure 2-14 Device 2: Monte Carlo graph of "Capacitance @ Vbias=0V" .....        | 23 |
| Figure 2-15 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V" ..... | 24 |
| Figure 2-16 Device 2: Monte Carlo graph of "Forward Current @ Vbias= 1.0V" ..... | 25 |
| Figure 2-17 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V" .....        | 26 |
| Figure 2-18 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V" ..... | 27 |
| Figure 2-19 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V" .....        | 28 |
| Figure 2-20 Device 2: Monte Carlo graph of "Capacitance @ Vbias=0V" .....        | 29 |
| Figure 2-21 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V" ..... | 30 |
| Figure 2-22 Device 2: Monte Carlo graph of "Forward Current @ Vbias= 1.0V" ..... | 31 |
| Figure 2-23 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V" .....        | 32 |
| Figure 2-24 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V" ..... | 33 |

# List of Tables

Table 1-1 Revision history .....5

Table 3-1 HSPICE warning message.....34

Table 3-2 Eldo warning message.....34

Table 3-3 Spectre warning message .....34

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# 1 Introduction

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## 1.1 Revision History

**Table 1-1 Revision history**

| Version | Date       | Author      | Remark                                                                                                                                                                                                                                                                       |
|---------|------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.1_p1  | 2011/04/20 | Yealing Lin | <ol style="list-style-type: none"><li>1. Original release.</li><li>2. This Monte Carlo model includes process variation characteristics extracted from model cards as below:<br/>l110ae_diode_v031.mdl<br/>l110ae_diode_v031.lib</li></ol>                                   |
| 0.2_p1  | 2011/10/05 | Laney Fan   | <ol style="list-style-type: none"><li>1. Modify the Monte Carlo model base on baseband model</li><li>2. This Monte Carlo model includes process variation characteristics extracted from model cards as below:<br/>l110ae_diode_v041.mdl<br/>l110ae_diode_v041.lib</li></ol> |
| 0.3_p1  | 2016/12/22 | Macro Chang | <ol style="list-style-type: none"><li>1. Modify the Monte Carlo model base on baseband model</li><li>2. This Monte Carlo model includes process variation characteristics extracted from model cards as below:<br/>l110ae_diode_v071.mdl<br/>l110ae_diode_v071.lib</li></ol> |
| 0.4_p1  | 2017/10/20 | Macro Chang | <ol style="list-style-type: none"><li>1. Modify the Monte Carlo model base on baseband model</li><li>2. This Monte Carlo model includes process variation characteristics extracted from model cards as below:<br/>l110ae_diode_v081.mdl<br/>l110ae_diode_v081.lib</li></ol> |

## 1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED\_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

## 1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below, and the simulation results from other versions of the simulators might be different

Synopsys HSPICE release 2014.9SP2  
Cadence SPECTRE version 15.1.0.644.isr13 64bit  
Mentor Graphics ELDO version 15.4

### 1.3.1 The combination of statistical model & corner model

The Monte-Carlo model and corner model (3 corners) are combined into a single model library. The user can choose either the Monte-Carlo model or a corner case by selecting different model sections for simulation. The following sections are included:

MC: Monte-Carlo diode model  
TT: Typical diode model  
FF: Fast diode model  
SS: Slow diode model

### 1.3.2 Sigma number for Monte-Carlo model

The user can define standard deviation at sigma level for Monte-Carlo simulation. If a larger sigma number is used, it means the model parameters will have greater variation. Sigma=3 is recommended to be used.

Follow these steps to invoke the model file during simulation.

Define standard deviation at the sigma level and pick up a value (1~3) of sigma to run MC simulations;

Example: `.param sigma=3`

Define the flag of Monte-Carlo simulation;

PROCESS: 1 to turn on the MC process simulation

0 to turn off the MC process simulation

Example: `.param process=1`

`.param process=0`

Define the number of Monte Carlo iterations and control MC simulations:

Example: `sweep monte=1000` (for HSPICE & ELDO)

`montecarlo numruns=1000` (for SPECTRE)

#### For Synopsys HSPICE

Copy following files into your simulation directory:

ModelFileName\_mc.mdl

ModelFileName\_mc\_statistical\_p.mdl

ModelFileName\_mc.lib

ModelFileName\_mc\_corner.lib

- Add the following statement in the input file to the simulator for 90% dimension Shrinkage.

`.Option scale=0.9`

`.lib "/ModelFileName_mc_corner.lib" CaseName`

, where CaseName could be mc, tt, ff, ss.

- Define device condition by 'Xxx' statement for sub-circuit format model, or

'Dxx' statement for non sub-circuit format model

Example1:(for sub-circuit format model)

`X1 Va Vc DeviceName AREA=1E-10 PJ=4E-5`

Example2:(for non sub-circuit format model)

`D1 Va Vc DeviceName AREA=1E-10 PJ=4E-5`

## **For Cadence SPECTRE**

Copy following files into your simulation directory:

ModelFileName\_mc.mdl.scs

ModelFileName\_mc\_statistical\_p.mdl.scs

ModelFileName\_mc.lib.scs

ModelFileName\_mc\_corner.lib.scs

- Add the following statement in the input file to the simulator for 90% dimension Shrinkage.

SetOption1 options scale=0.9

include "./ModelFileName\_mc\_corner.lib.scs" section=CaseName

, where CaseName could be mc, tt, ff, ss.

- Define device condition by 'Xxx' statement for sub-circuit format model, or

'Dxx' statement for non sub-circuit format model

Example1:(for sub-circuit format model)

x1 (Va Vc) DeviceName area=1e-10 pj=4e-5

Example2:(for non sub-circuit format model)

d1 Va Vc DeviceName area=1e-10 pj=4e-5

## **For Mentor Graphics ELDO**

Copy following files into your simulation directory:

ModelFileName\_mc.mdl.eldo

ModelFileName\_mc\_statistical\_p.mdl.eldo

ModelFileName\_mc.lib.eldo

ModelFileName\_mc\_corner.lib.eldo

- Add the following statement in the input file to the simulator for 90% dimension Shrinkage.

.Option scale=0.9

.lib "./ModelFileName\_mc\_corner.lib.eldo" CaseName

, where CaseName could be mc, tt, ff, ss.

- Define device condition by 'Xxx' statement for sub-circuit format model, or

'Dxx' statement for non sub-circuit format model

Example1:(for sub-circuit format model)

X1 Va Vc DeviceName AREA=1E-10 PJ=4E-5

Example2:(for non sub-circuit format model)

D1 Va Vc DeviceName AREA=1E-10 PJ=4E-5

ModelFileName: l110ae\_diode\_v041

DeviceName: dion\_l110e, diop\_l110e, dion\_33\_l110e, diop\_33\_l110e, dionw\_l110e, diodnw\_l110e and diopw\_l110e



As stated above that one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

**Geometry range(Before 90% shrinkage):**

HS1.2V DION  
 $0.5\mu\text{m} \leq \text{WDES} \leq 14.3\mu\text{m}$   
 $5\mu\text{m} \leq \text{LDES} \leq 25\mu\text{m}$

HS1.2V DIOP  
 $0.5\mu\text{m} \leq \text{WDES} \leq 14.3\mu\text{m}$   
 $5\mu\text{m} \leq \text{LDES} \leq 25\mu\text{m}$

IO 3.3V DION  
 $0.5\mu\text{m} \leq \text{WDES} \leq 14.3\mu\text{m}$   
 $5\mu\text{m} \leq \text{LDES} \leq 25\mu\text{m}$

IO 3.3V DIOP  
 $0.5\mu\text{m} \leq \text{WDES} \leq 14.3\mu\text{m}$   
 $5\mu\text{m} \leq \text{LDES} \leq 25\mu\text{m}$

DIONW  
Area =  $1.77\text{E-}7 \text{ m}^2$   
 $\text{Pj} = 4.93\text{E-}3 \text{ m}$

DIODNW  
Area =  $3.2\text{E-}8 \text{ m}^2$   
 $3.2\text{E-}4 \text{ m} \leq \text{Pj} \leq 6.4\text{E-}3 \text{ m}$

DIOPW  
Area =  $3.2\text{E-}8 \text{ m}^2$   
 $\text{Pj} = 3.2\text{E-}4 \text{ m}$

**Temperature range:**

$-50\text{ }^\circ\text{C} \sim +125\text{ }^\circ\text{C}$

## 2 Monte Carlo Simulation Results for Process Variation

Monte Carlo iterations is 1000 times to be performed in following simulation condition and the following corner points show TT, FF, and SS value, individually. Reasonable Monte Carlo iterative times are 30. The statistical significance of 30 iterations is quite high. If the circuit operates correctly for all 30 iterations, there is 99% probability that over 80% of all possible component values operate correctly. The relative error of a quantity determined through Monte Carlo analysis is proportional to (iteration times)<sup>(-1/2)</sup>.

### 2.1 Simulation result plots

The following figures show out DIODE device's process variation at 25 C. Area/Pj dimension is before shrinkage.

#### 2.1.1 1.2V DION simulation results:

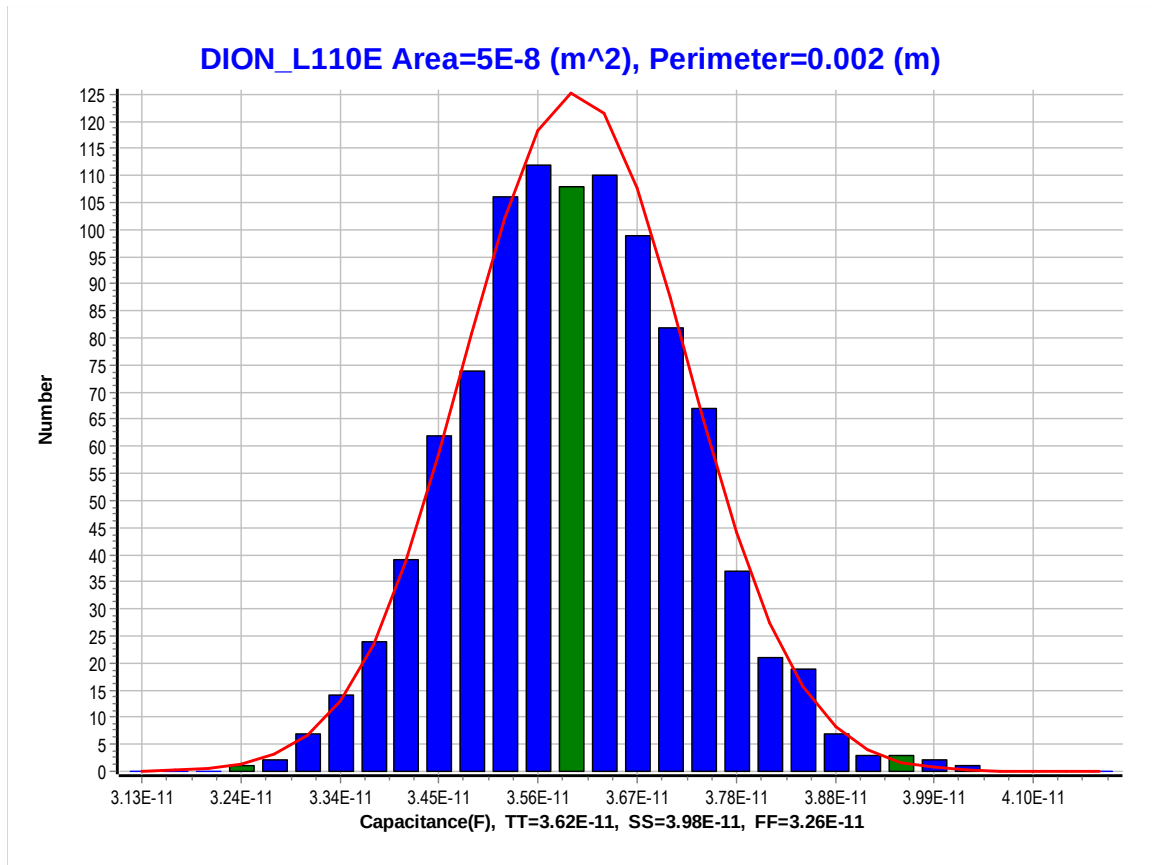
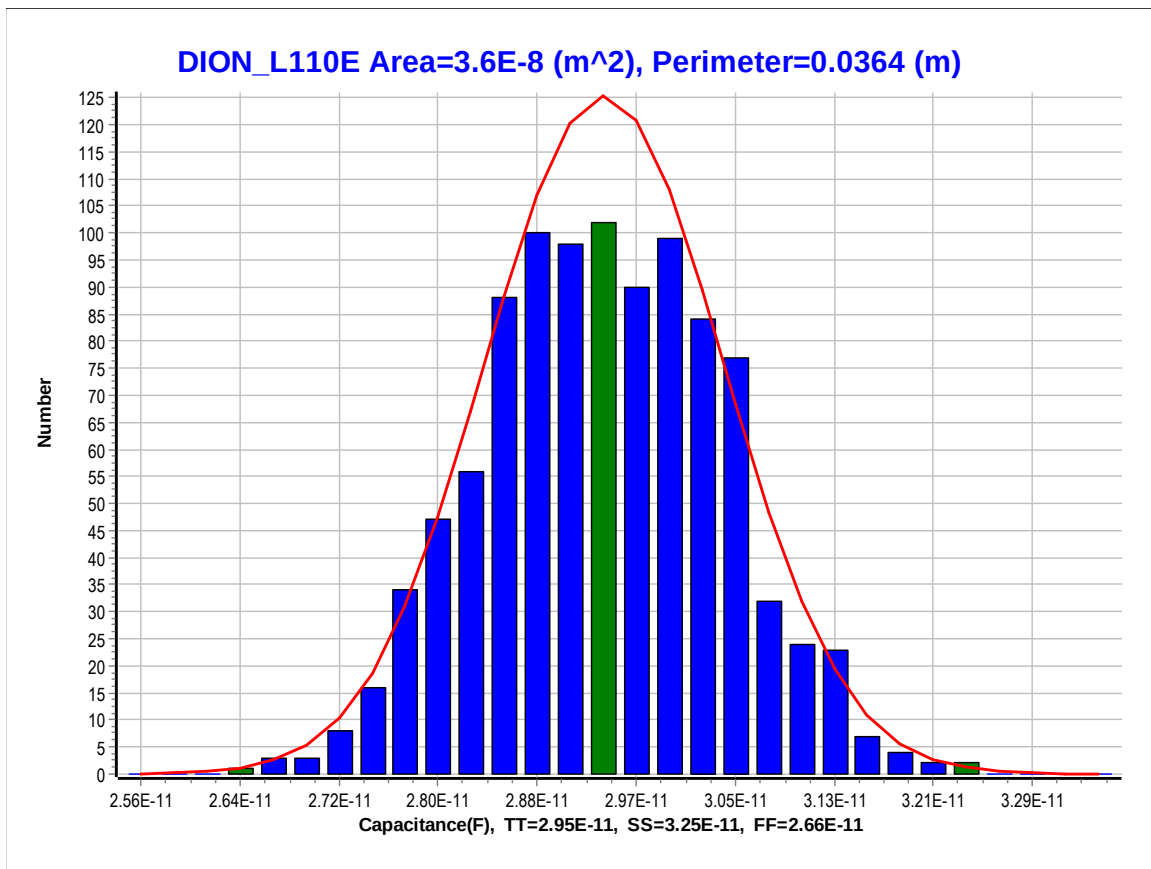
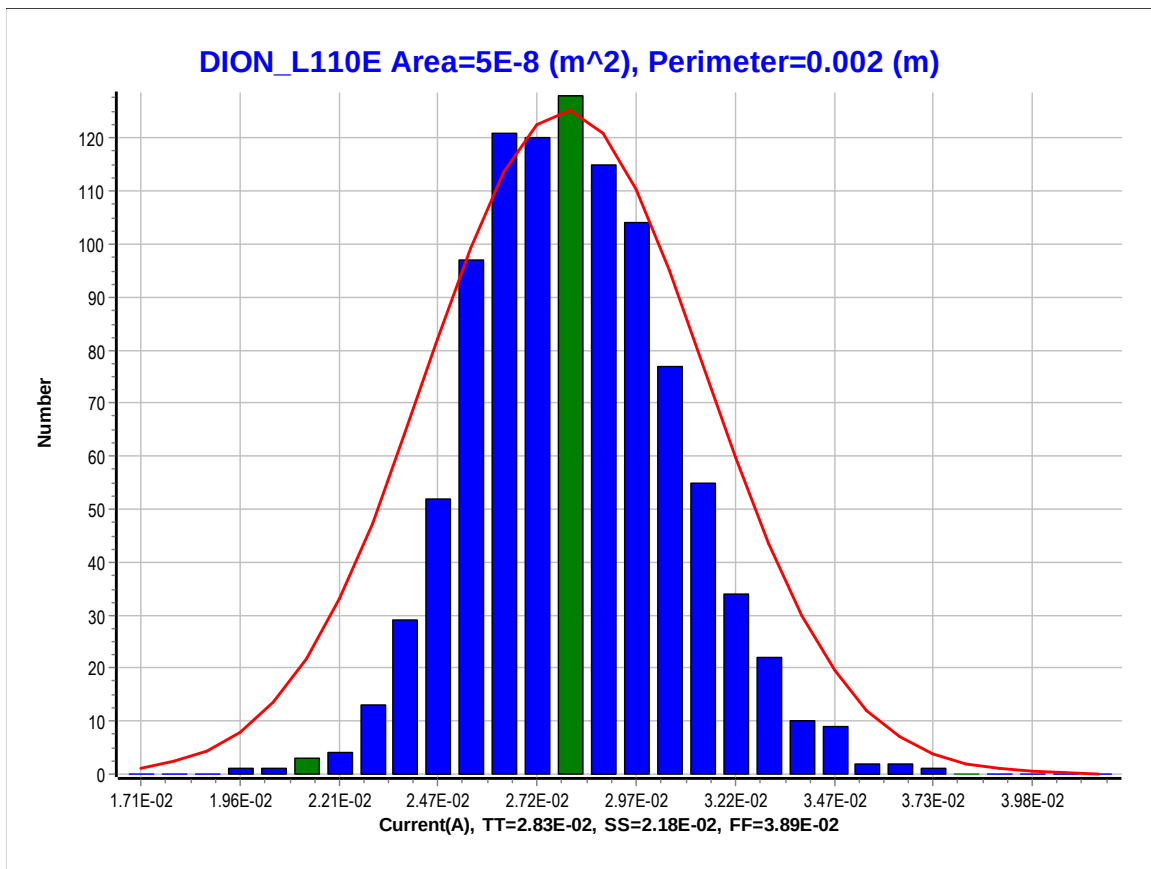


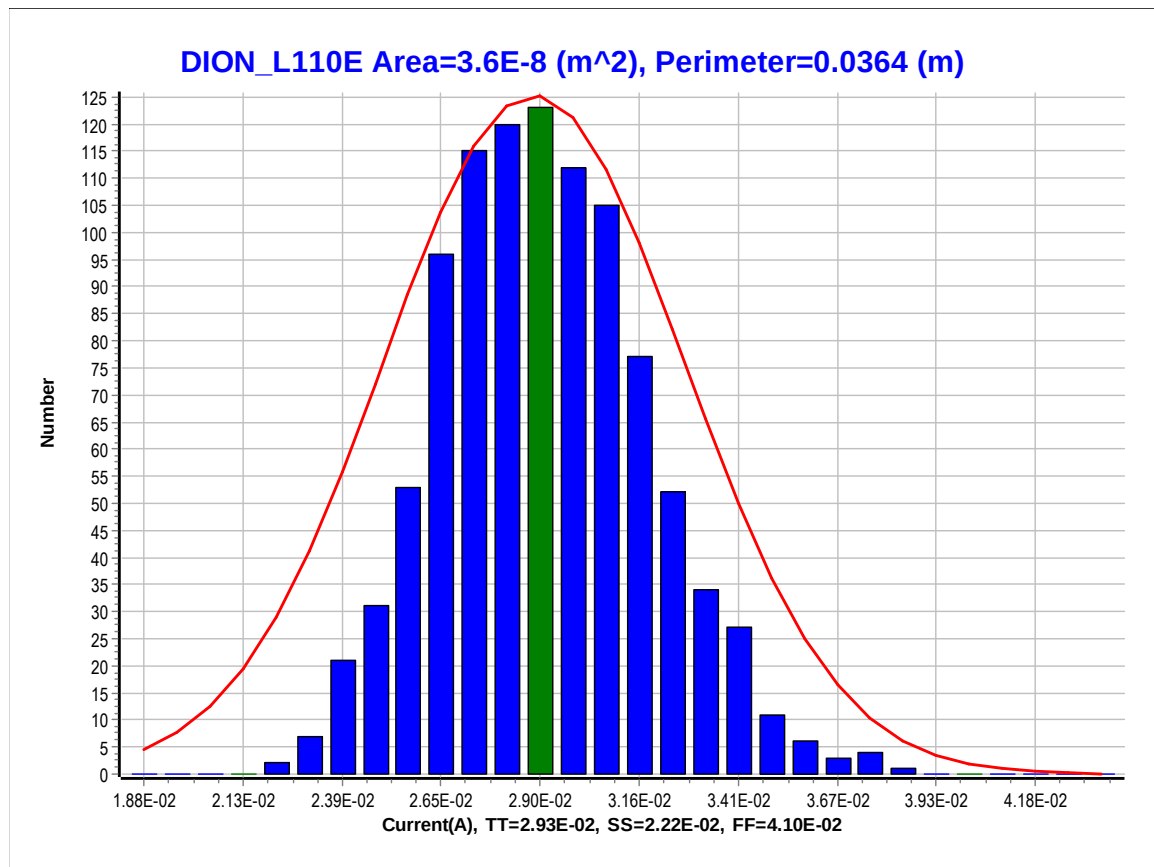
Figure 2-1 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V"



**Figure 2-2 Device 2: Monte Carlo graph of "Capacitance @ Vbias=0V"**

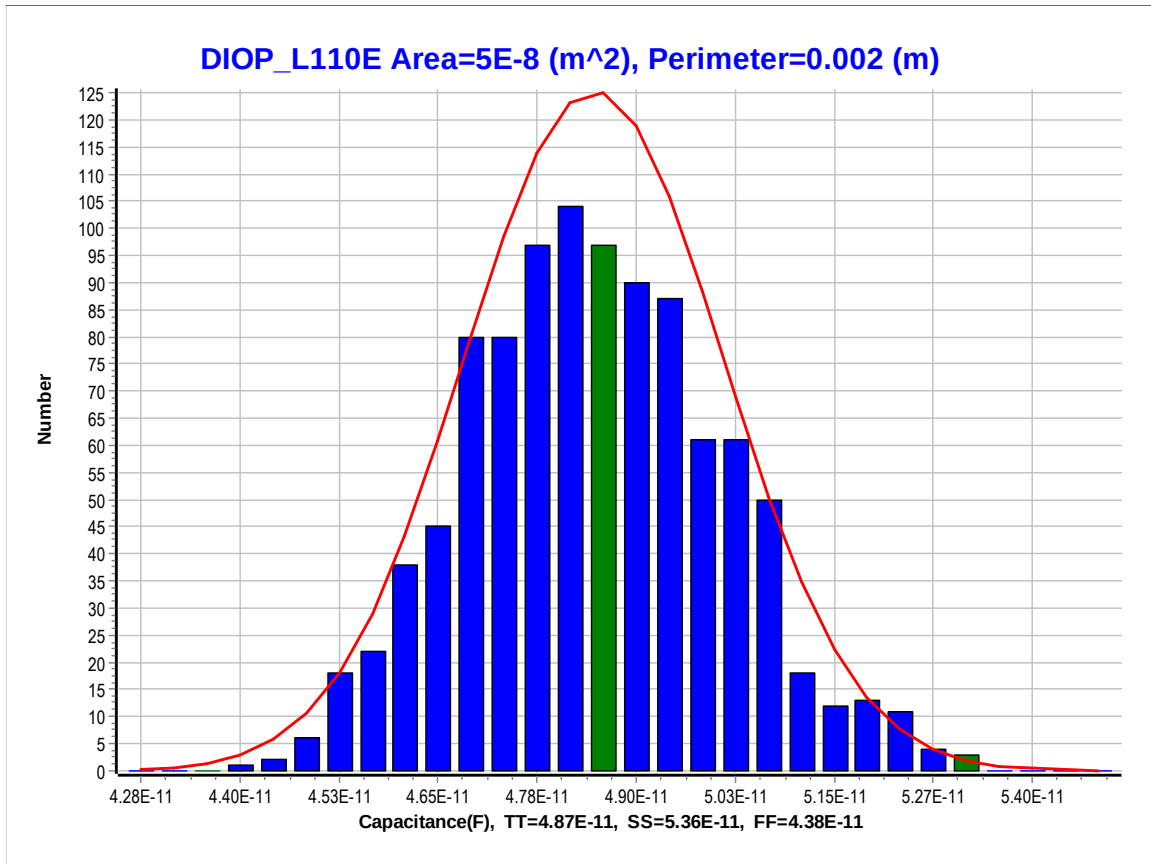


**Figure 2-3 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"**

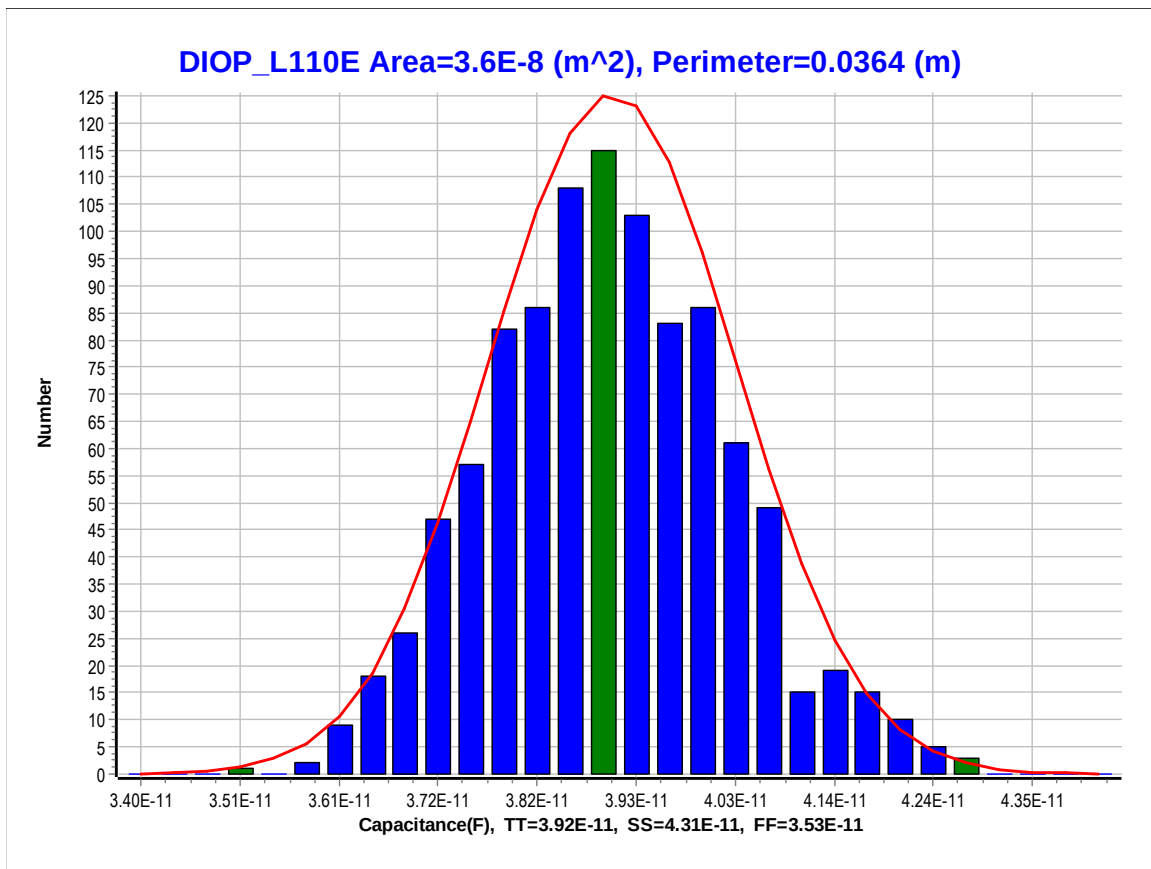


**Figure 2-4 Device 2: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"**

## 2.1.2 1.2V DIOP simulation results:



**Figure 2-5 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V"**



**Figure 2-6 Device 2: Monte Carlo graph of "Capacitance @ Vbias=0V"**

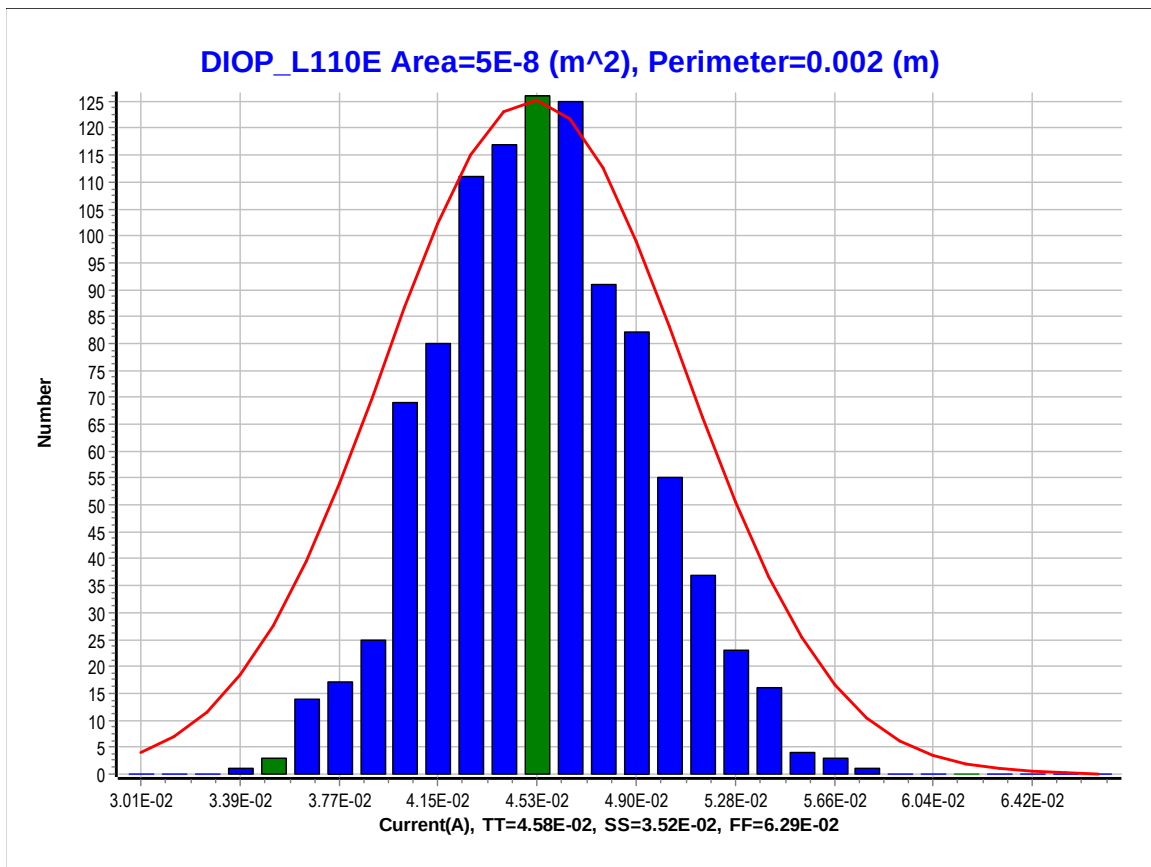


Figure 2-7 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"



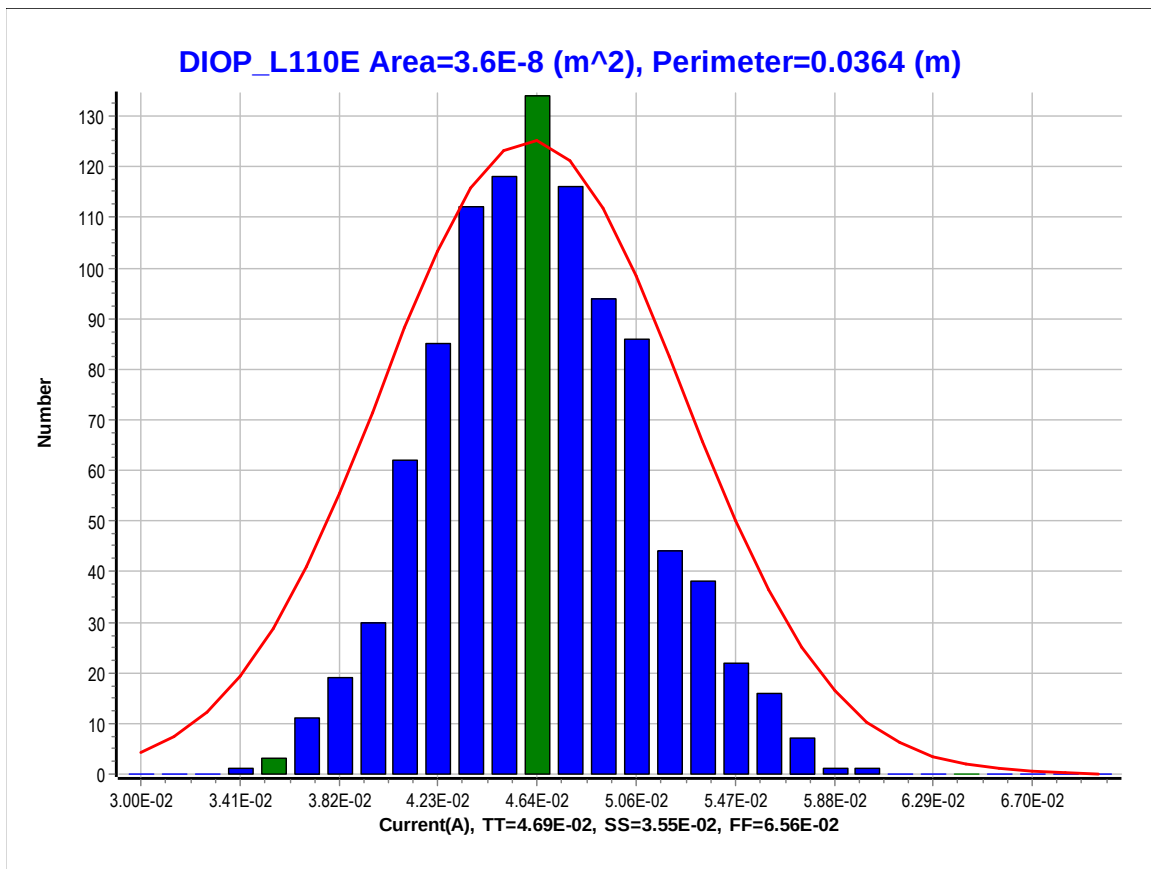
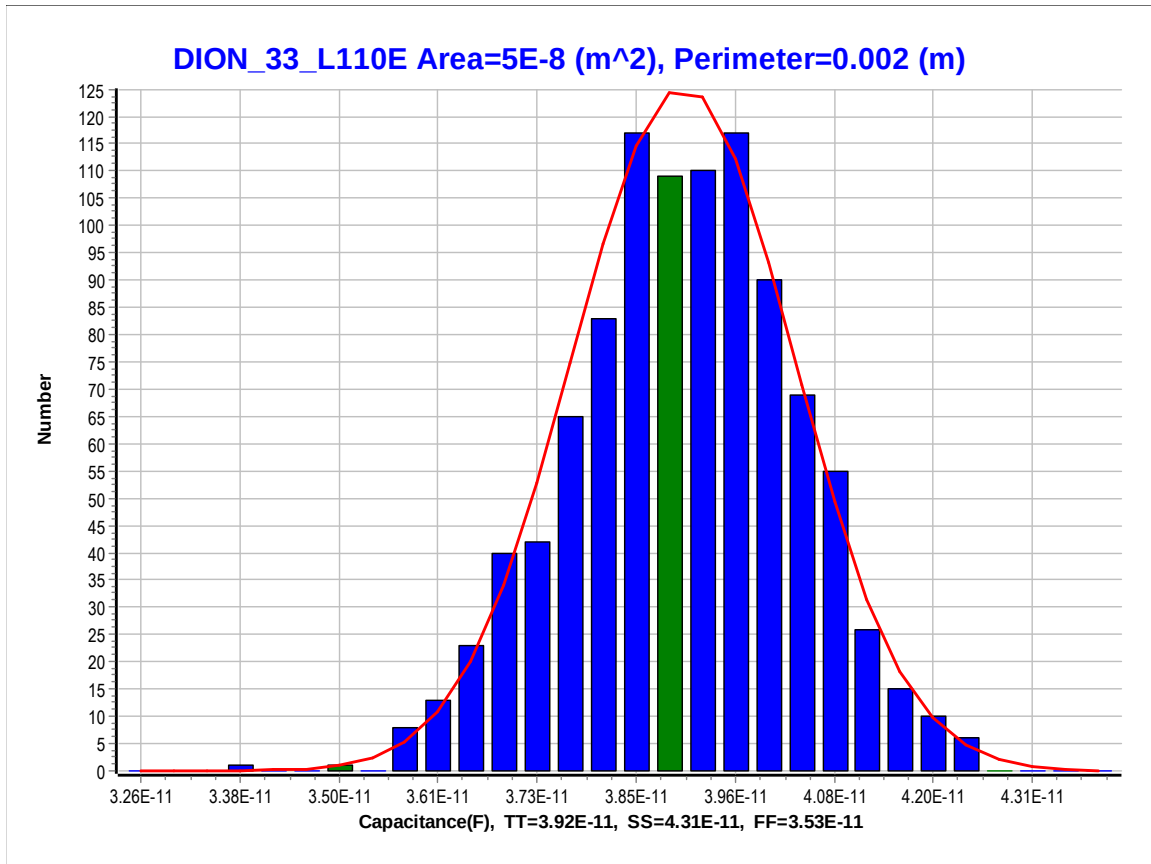
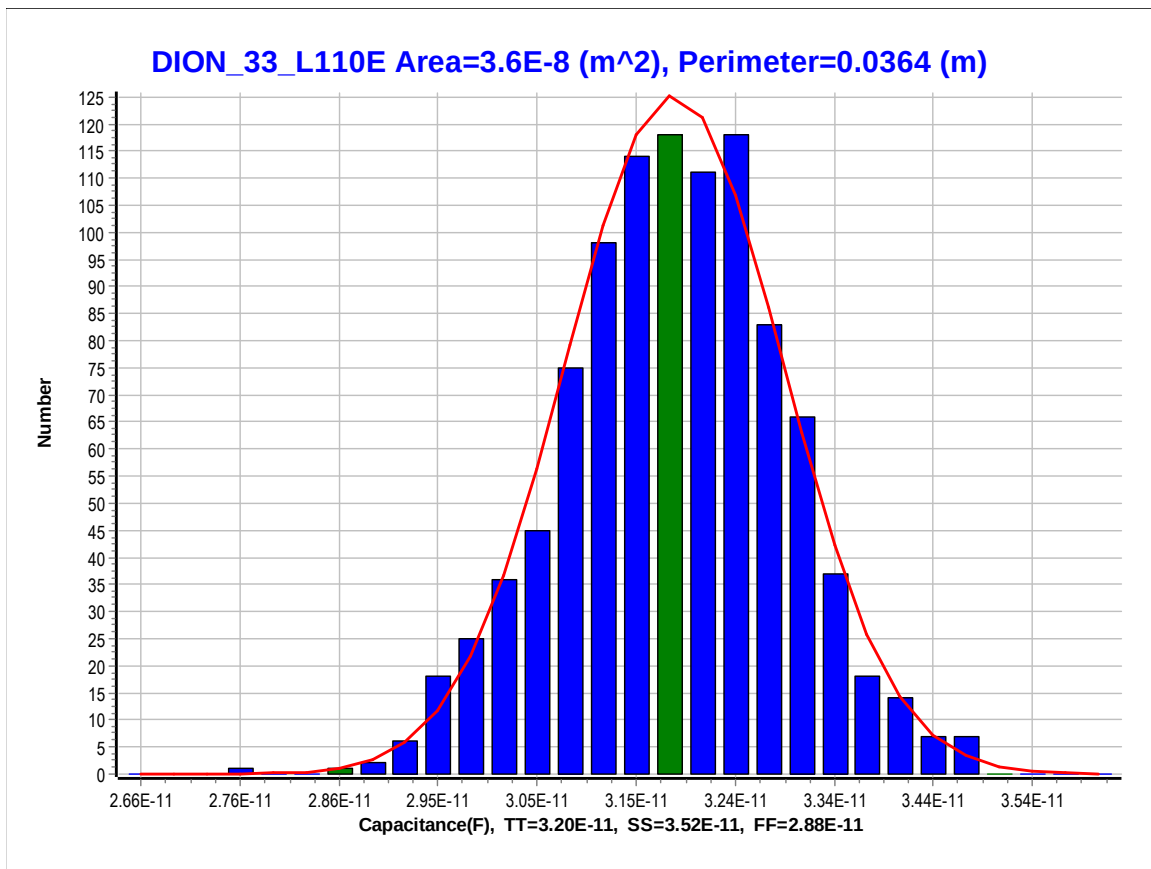


Figure 2-8 Device 2: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"

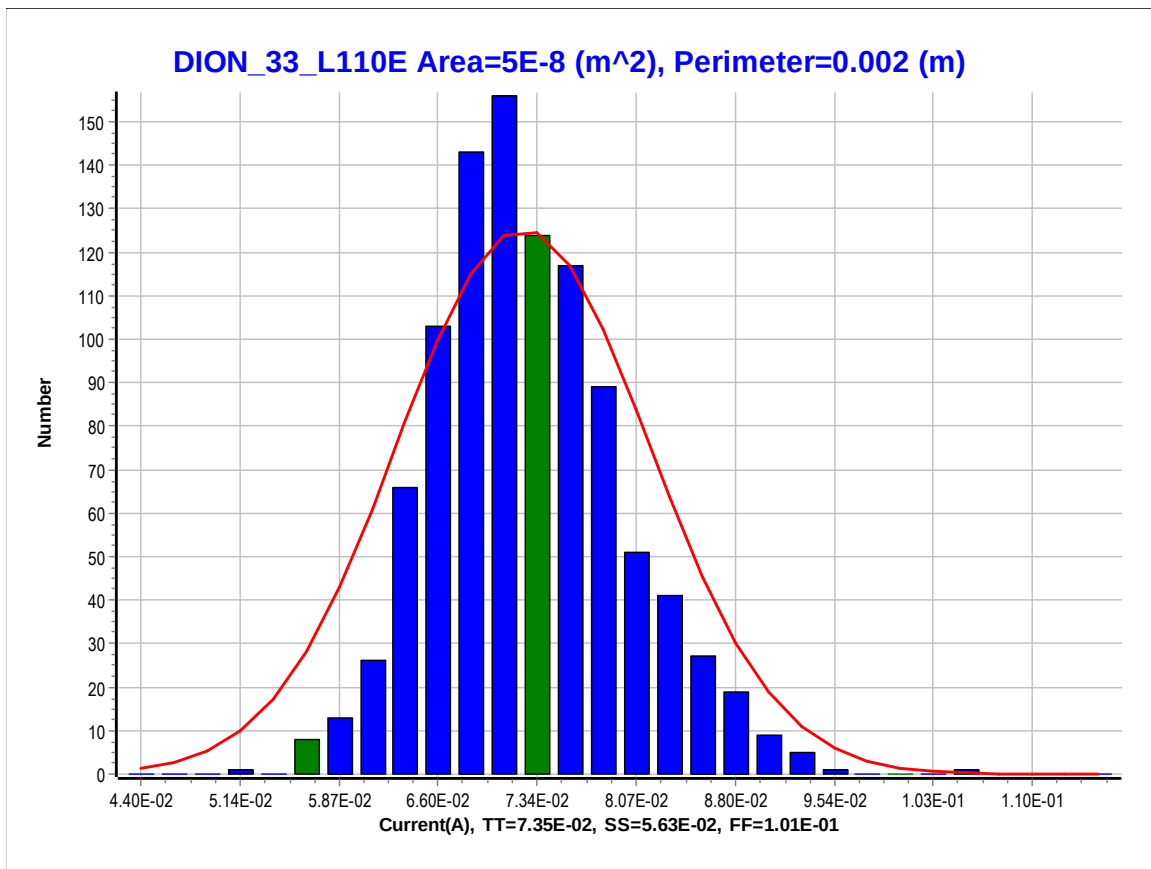
### 2.1.3 3.3V DION simulation results:



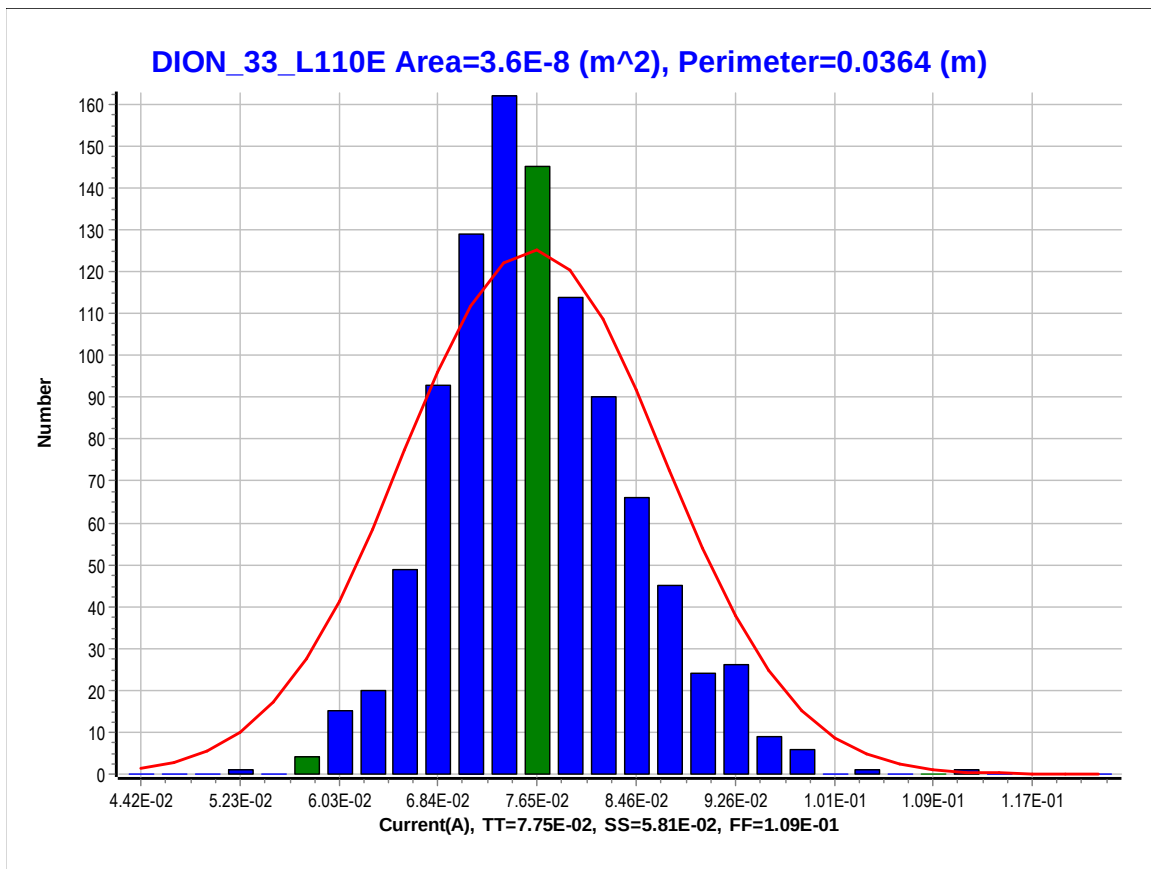
**Figure 2-9 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V"**



**Figure 2-10 Device 2: Monte Carlo graph of "Capacitance @ Vbias=0V"**

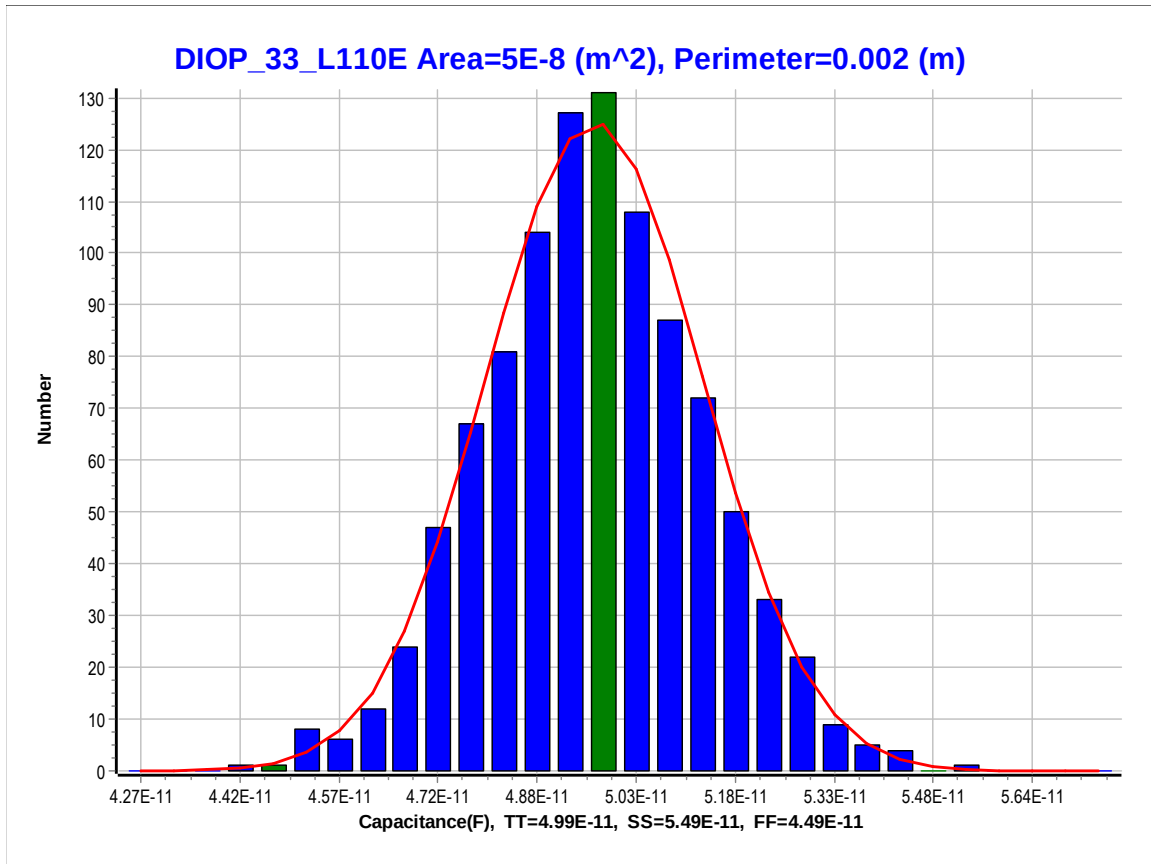


**Figure 2-11 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"**

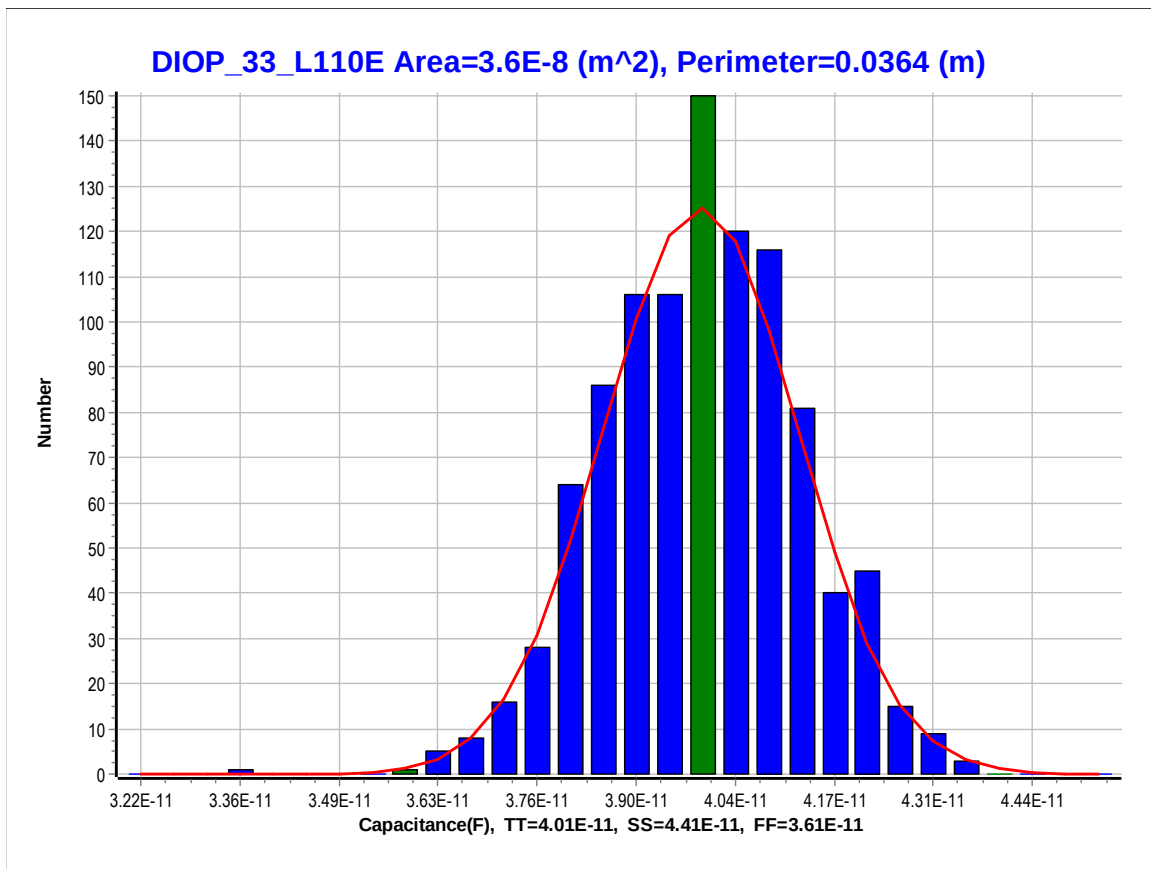


**Figure 2-12 Device 2: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"**

## 2.1.4 3.3V DIOP simulation results:



**Figure 2-13 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V"**



**Figure 2-14 Device 2: Monte Carlo graph of "Capacitance @ Vbias=0V"**

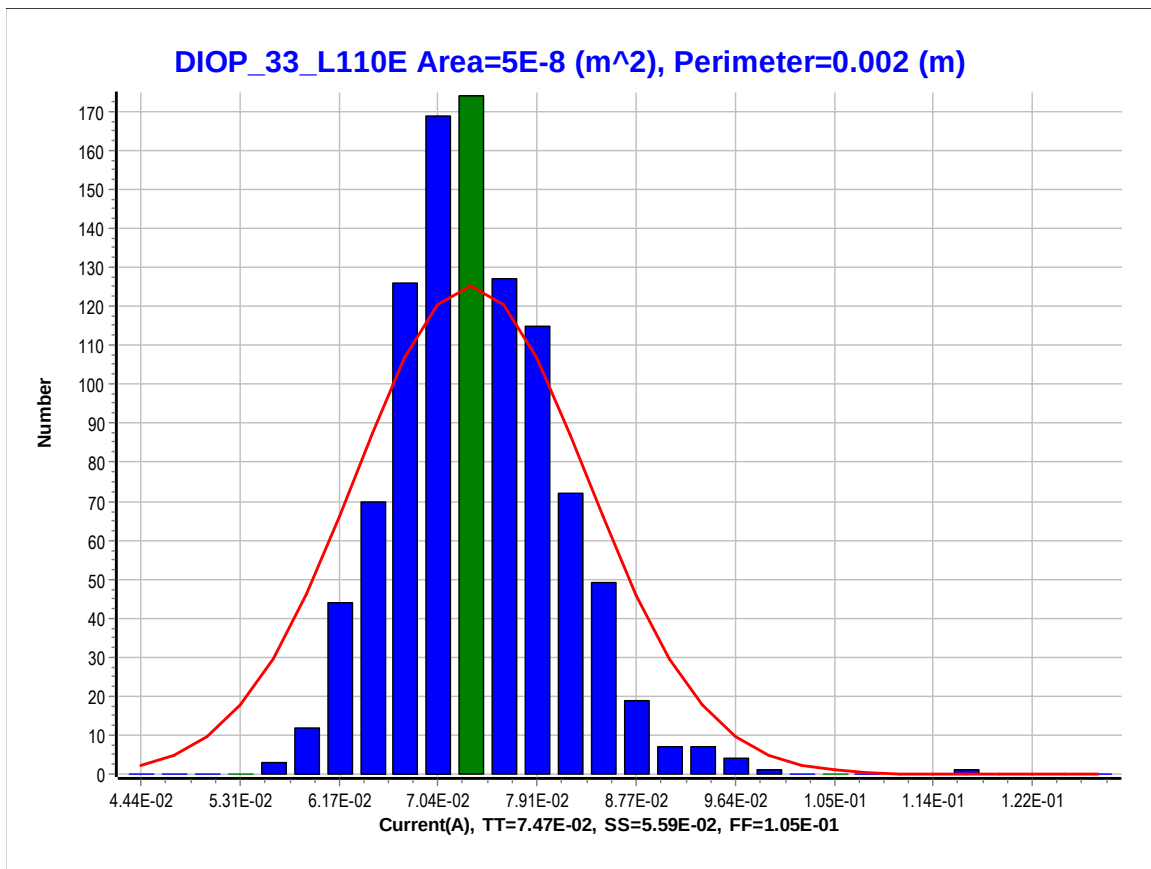


Figure 2-15 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"



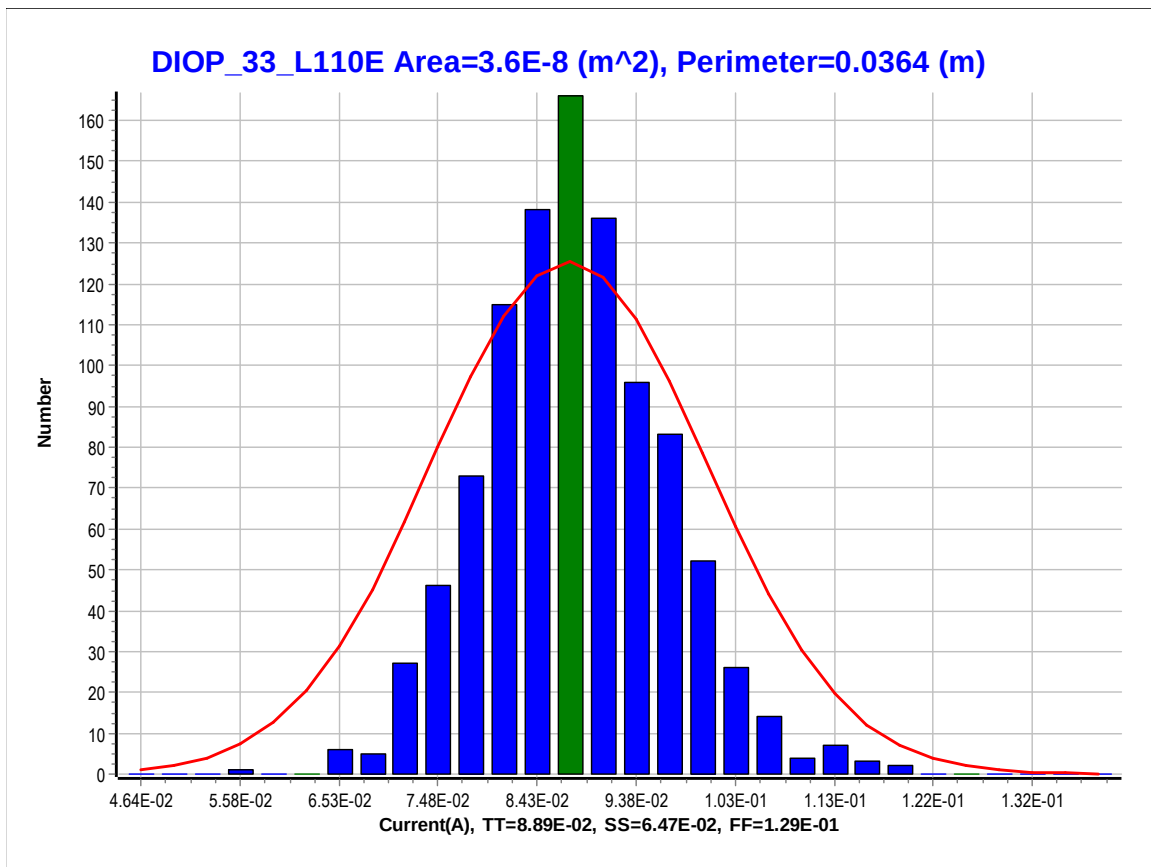
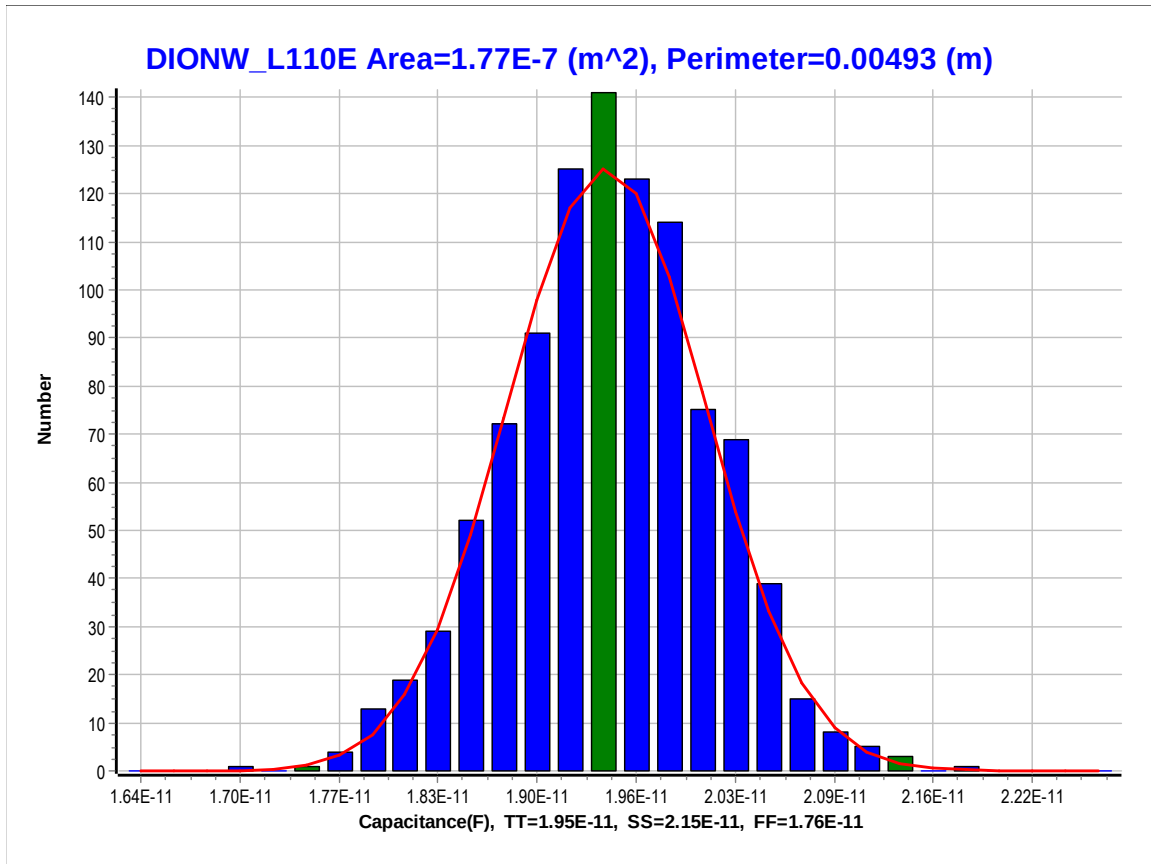
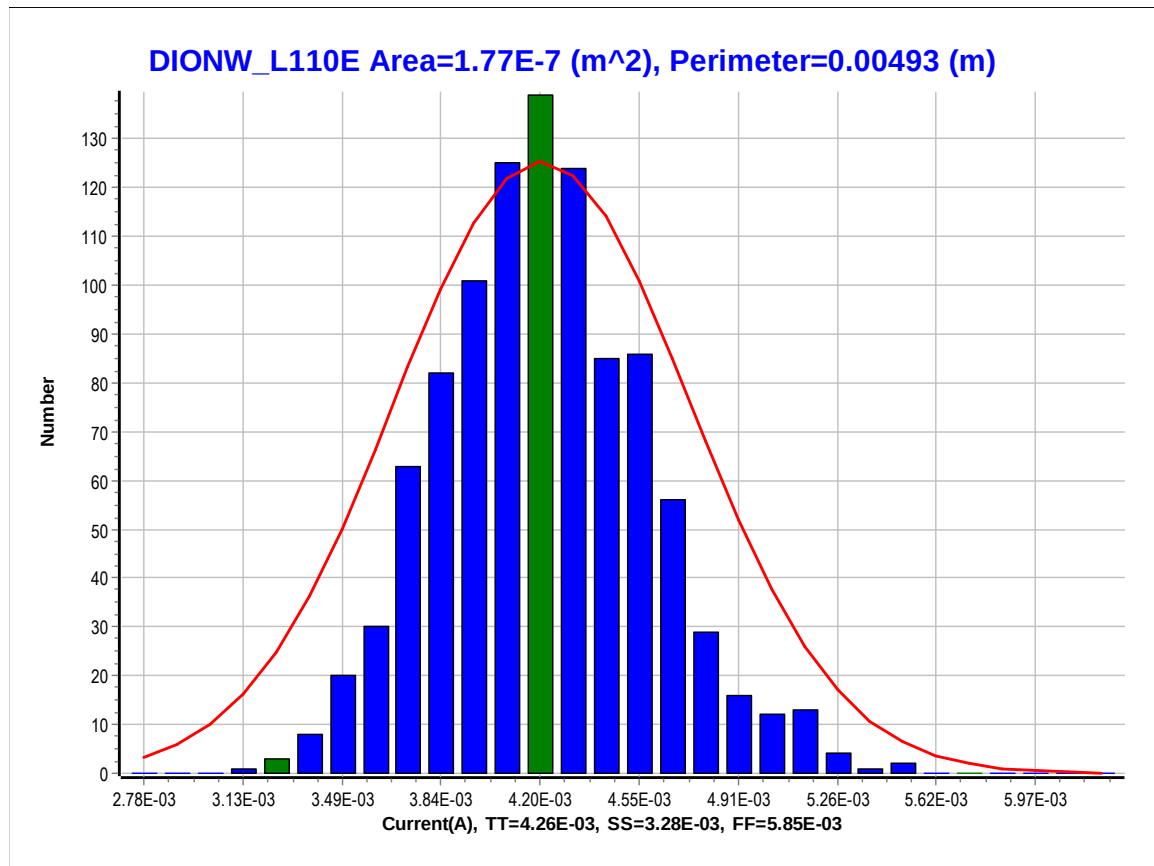


Figure 2-16 Device 2: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"

## 2.1.5 1.2V DIONW simulation results:

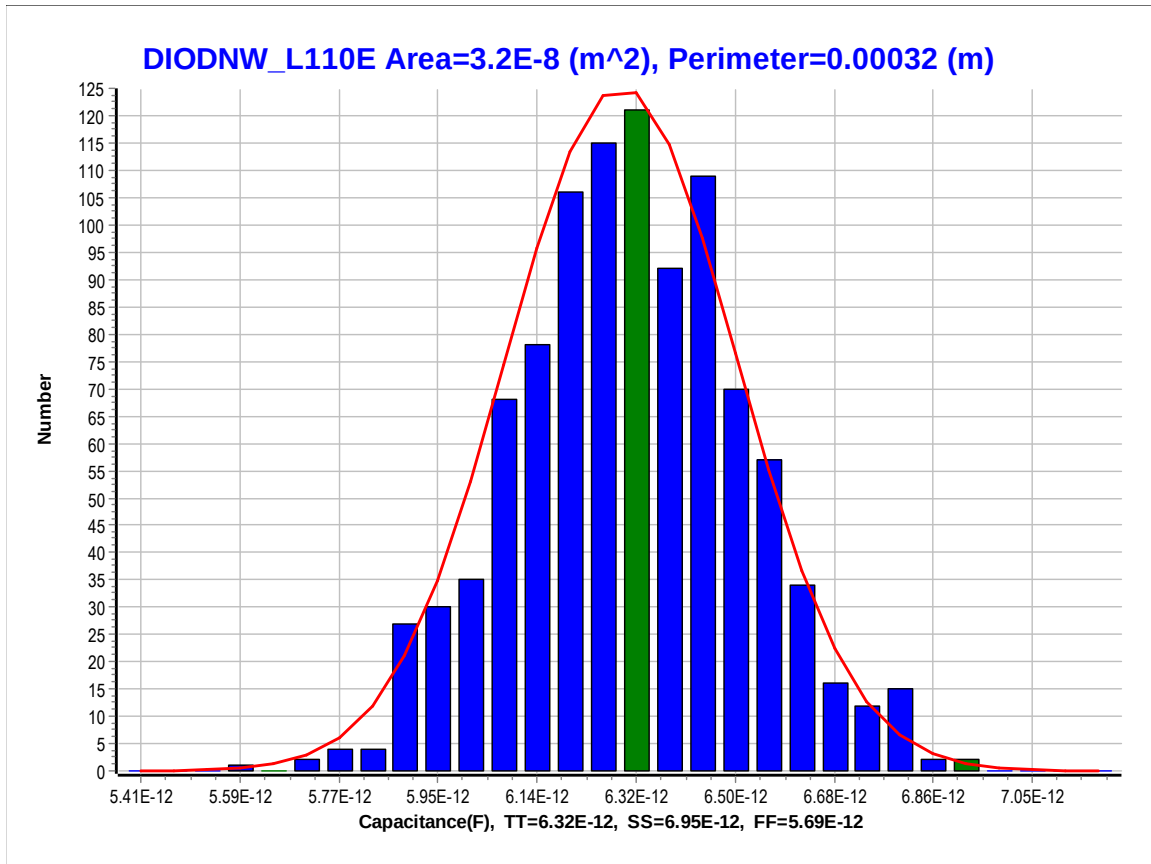


**Figure 2-17 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V"**

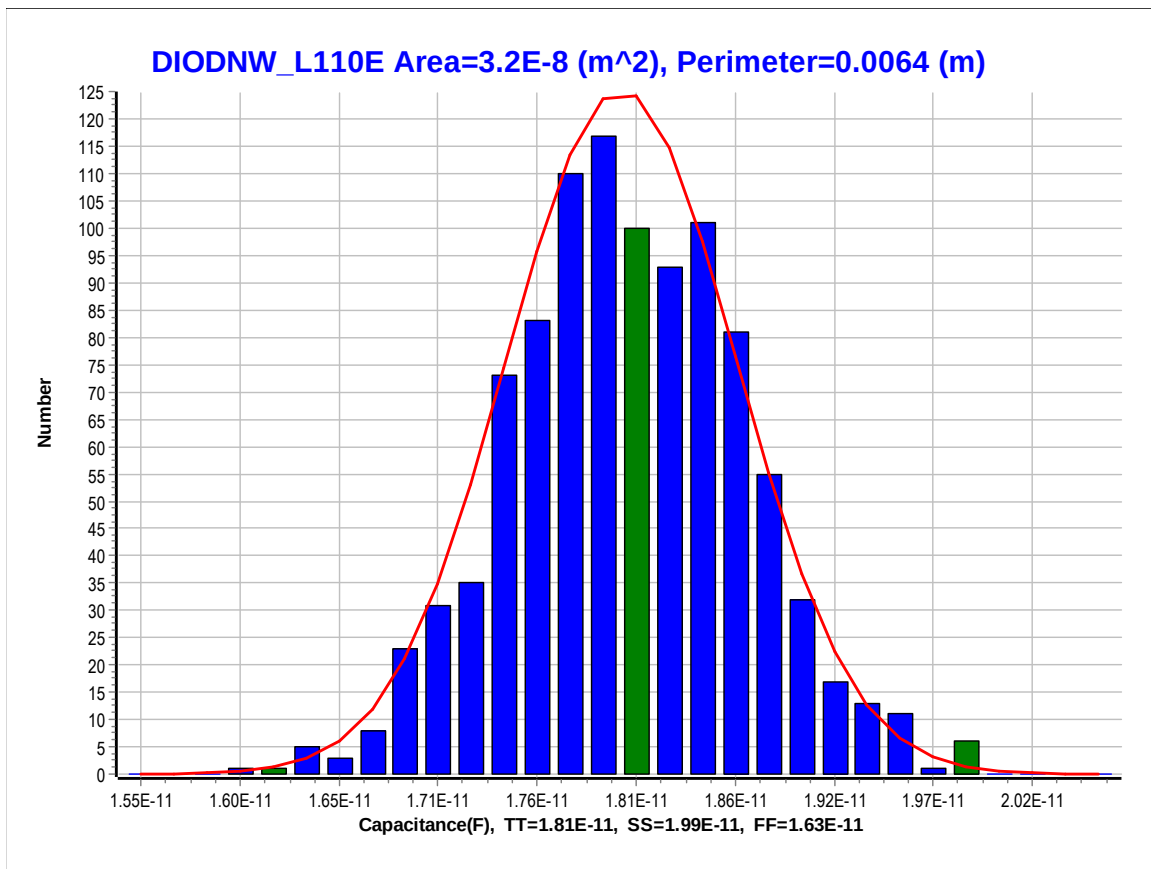


**Figure 2-18 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"**

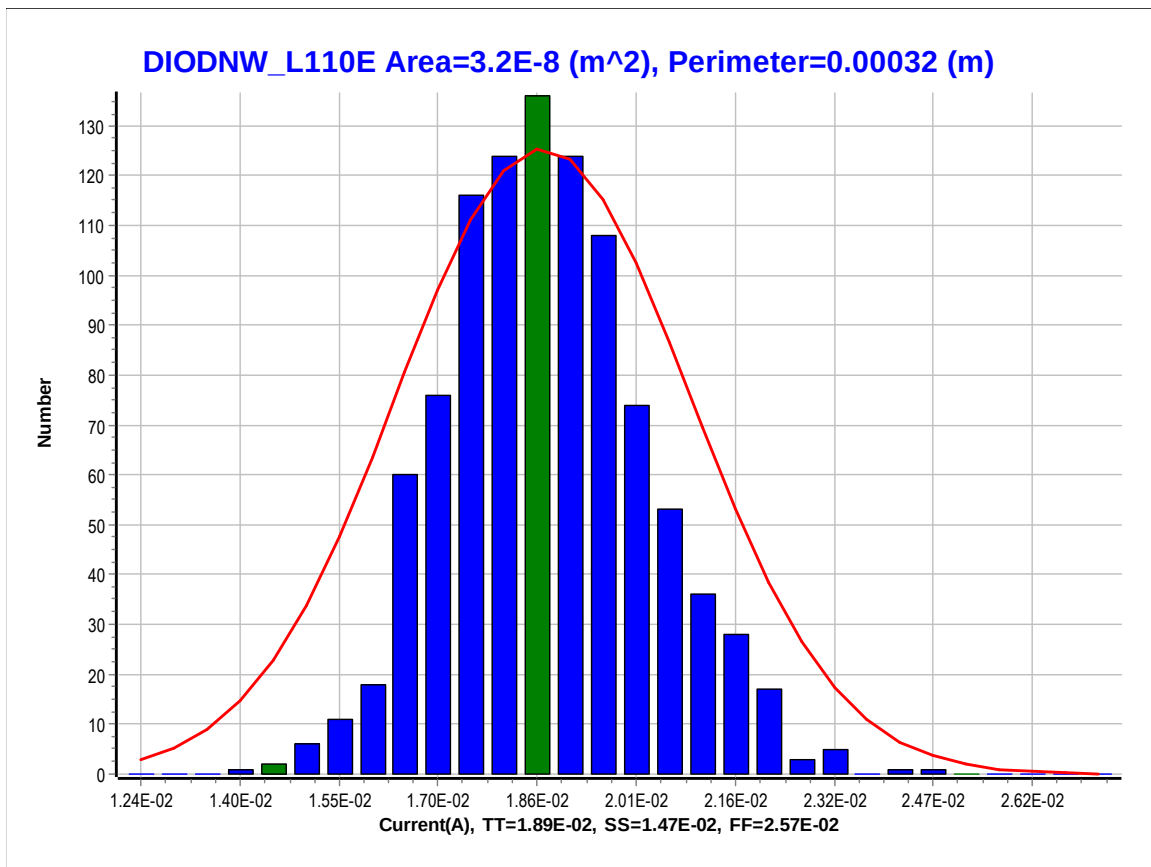
## 2.1.6 1.2V DIODNW simulation results:



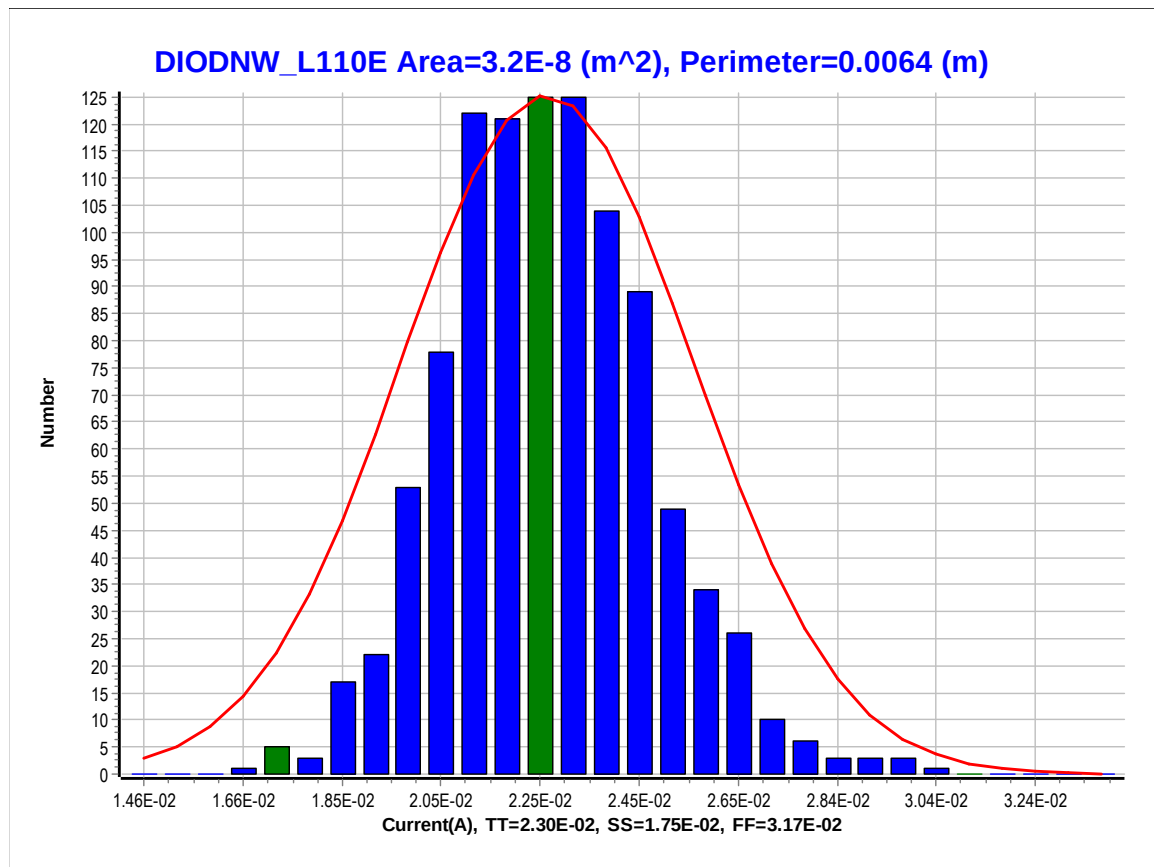
**Figure 2-199 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V"**



**Figure 2-20 Device 2: Monte Carlo graph of "Capacitance @ Vbias=0V"**

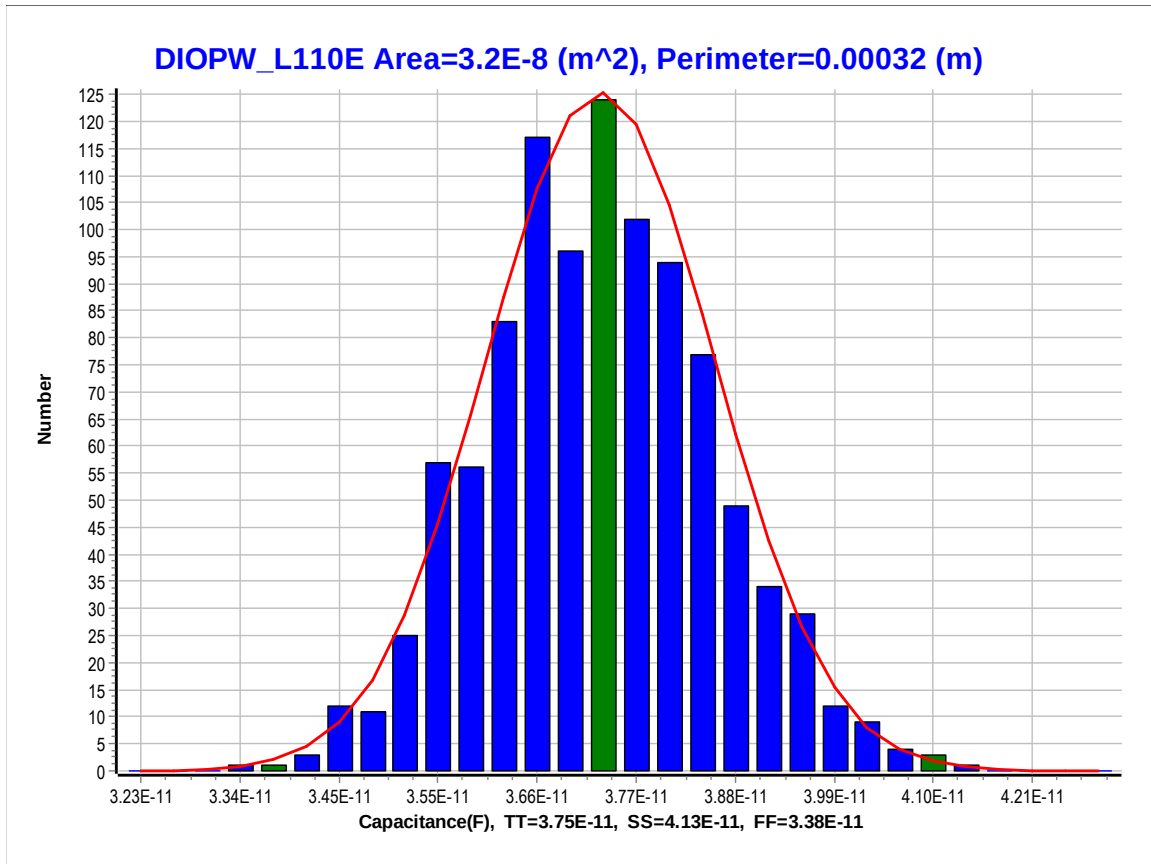


**Figure 2-21 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"**



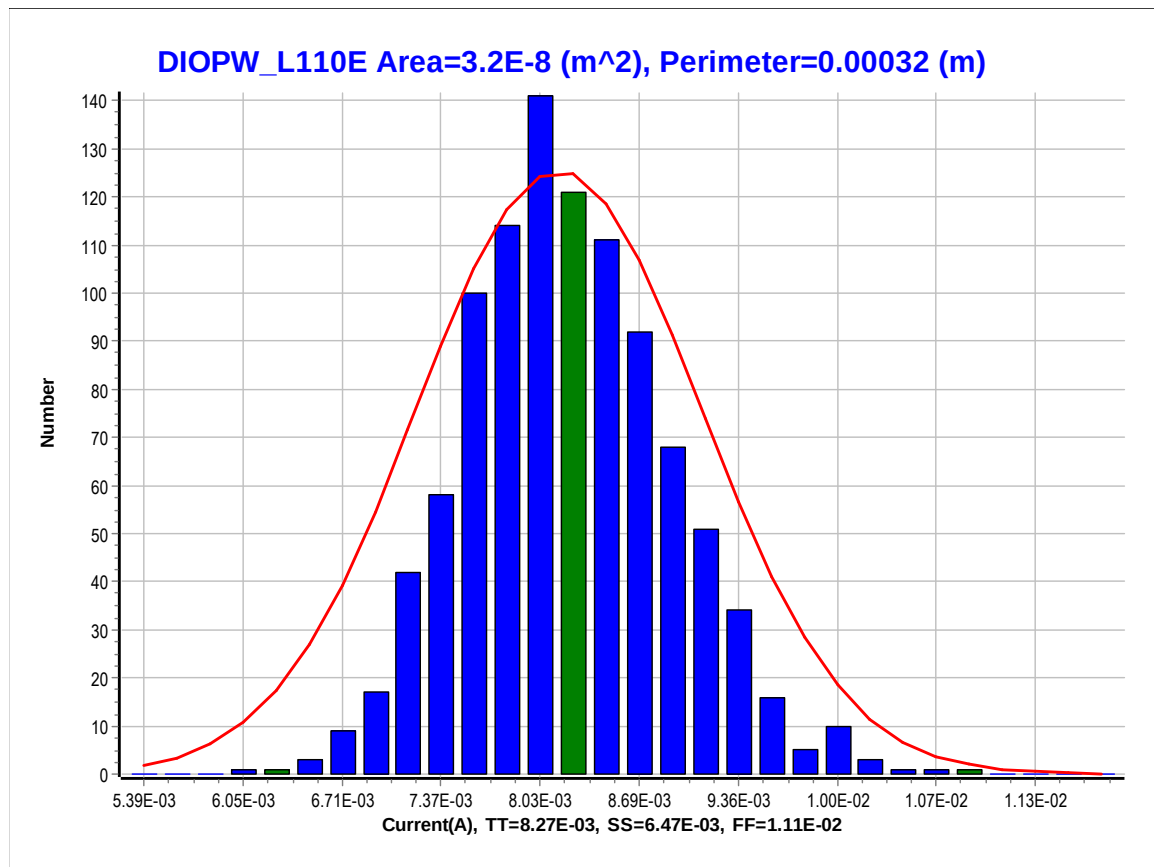
**Figure 2-22 Device 2: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"**

## 2.1.7 1.2V DIOPW simulation results:



**Figure 2-23 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V"**





**Figure 2-24 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"**

---

## 3 Appendix

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### 3.1 HSPICE Warning Message

**Table 3-1 HSPICE warning message**

| NO. | Warning Message | Explanation |
|-----|-----------------|-------------|
| 1.  | None            | None        |

### 3.2 ELDO Warning Message

**Table 3-2 Eldo warning message**

| NO. | Warning Message | Explanation |
|-----|-----------------|-------------|
| 1.  | None            | None        |

### 3.3 SPECTRE Warning Message

**Table 3-3 Spectre warning message**

| NO. | Warning Message | Explanation |
|-----|-----------------|-------------|
| 1.  | None            | None        |